6 A railway depot uses an electrical de-icing unit to prevent ice from forming on a track-switch mechanism. The unit contains a resistive heating element supplied by an ideal step-down transformer. A diagram of the setup is shown in Fig. 6.1 below.

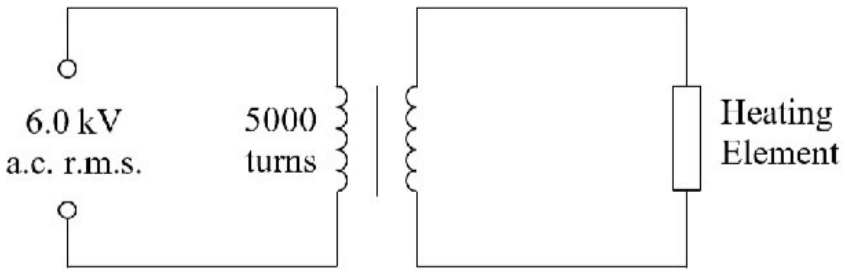


Fig. 6.1

The primary coil of the transformer is connected to a 6.0 kV, 50 Hz sinusoidal alternating supply. The potential difference across the secondary coil is 240 V r.m.s.. The primary coil has 5000 turns.

(a) Determine the number of turns on the secondary coil.

number of turns= ..... [1]

(b) The heating element behaves as a pure resistor and dissipates a mean power of 4.6 kW. The instantaneous current I in the heating element is represented by

$$I = I_o \sin \omega t \quad .$$

Determine I_o and ω.

I_o = ... A

ω = ... rad s^{-1} [3]

(c) Determine the r.m.s. current in the primary coil of the transformer.

current = ... A [2]

(d) On Fig. 6.2, sketch the variation with time t of the power P dissipated in the resistor from $t = 0$ to $t = 0.020$ s.

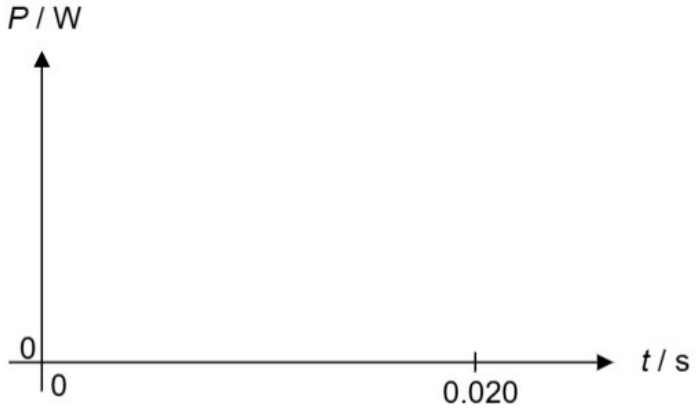


Fig. 6.2

[2]

(e) On Fig. 6.3, sketch the variation of the net amount of charge Q which flowed past a point in the resistor from $t = 0$ to $t = 0.020$ s. (Numerical values for the charge are not expected).

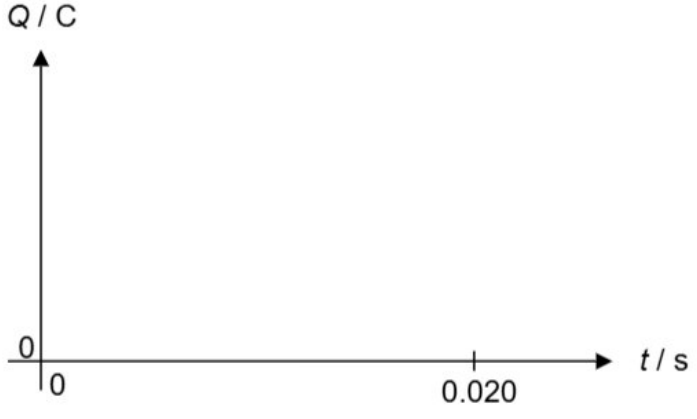


Fig. 6.3

[1]

[Total: 9]